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Original Article

Benchmarking the MARS code for molten salt reactor applications using MSRE transient experiments

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ABSTRACT

The MARS thermal-hydraulic system code was extended and assessed for applications to liquid-fueled molten salt reactors (MSRs) using transient experiments from the Molten Salt Reactor Experiment (MSRE). Molten salt thermophysical property models were implemented in the MARS code first. To apply the point kinetics model to an MSR, a delayed neutron precursor (DNP) transport equation was incorporated in place of the conventional precursor balance model. A transport-based decay heat model was also developed.

In this study, the extended MARS code was validated against three MSRE transients: pump startup, pump coastdown, and natural circulation tests. The first two tests, performed under near-zero-power conditions, separately examine DNP drift effects. MARS reproduced essential reactivity changes induced by fuel flow variations, including the oscillatory behavior of the pump startup test associated with fuel transit time and asymptotic reactivity loss. The natural circulation test involved coupled neutronic and thermal-hydraulic effects. While discrepancies in absolute fuel salt temperature values were observed due to uncertain initial conditions, the overall trends of fuel salt temperature and core power were reasonably predicted.

The results demonstrate that the extended MARS code captures key MSR-specific phenomena well. They also highlight the existing limitations and suggest directions for future code development.

1. Introduction

In recent years, molten salt reactors (MSRs) have regained significant attention as a promising advanced reactor concept owing to their inherent safety features, high operating temperature capability, and fuel cycle flexibility [1,2]. In MSRs, nuclear fuel is dissolved in a molten salt that simultaneously serves as the coolant, enabling low-pressure operation, efficient heat removal, and the potential for online fuel processing. These characteristics offer several advantages over conventional solid-fueled reactors, including improved thermal efficiency, enhanced passive safety, and reduced long-lived radioactive waste.

The unique characteristics of MSRs, however, pose substantial challenges for reactor design and safety analysis. Unlike solid-fueled reactors, the circulation of liquid fuel leads to stronger coupling between reactor kinetics and thermal-hydraulics [3]. In particular, delayed neutron precursors (DNPs) are transported by the flowing molten salt fuel, resulting in a loss of delayed neutrons from the reactor core. Decay heat generated by radioactive fission products is distributed throughout the reactor system. Accurate modeling of these phenomena is essential for reliable design and safety analyses of MSRs.

To address these challenges, considerable efforts have been devoted to the development of analysis codes for MSRs. Two major approaches can be identified. One approach is to develop new system analysis codes specifically tailored for MSR applications [4–7]. This approach offers high modeling fidelity but typically requires substantial development effort and extensive verification and validation. The other approach is to extend and modify existing thermal-hydraulic system codes, originally developed for solid-fueled reactors, to make them applicable to MSRs [8–12]. This approach benefits from the maturity, robustness, and extensive validation history of existing codes, but requires careful modifications to capture MSR-specific phenomena. Each approach has its own advantages and limitations.

In this context, the authors extended the MARS thermal-hydraulic system code for application to liquid-fueled MSRs. MARS is a well-established system code built on the consolidated framework of RELAP5 and COBRA-TF [13–15]. It adopts a two-fluid model to simulate two-phase flows in nuclear systems and has been extensively validated for water-cooled reactor applications. It is currently used for the design

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and safety analysis of nuclear systems as well as a regulatory audit calculation tool in Korea. The MARS code has also been extended for non-water-cooled reactors [16,17], such as sodium-cooled fast reactors and gas-cooled reactors. Recently, several key modifications have been implemented to enable the application of MARS to MSR analysis [18, 19]. First, thermophysical property models for various molten salts, such as LiF-BeF_2-ZrF_4-UF_4, LiF-BeF_2, KCl-UCl_3, and FUNaK, were incorporated as functions of temperature and pressure. Second, a modified point-kinetics model (PKM) was developed to account for the transport of DNPs in flowing liquid fuel. Third, a decay heat model that considers the transport of radioactive fission products was implemented to represent distributed heat generation throughout the reactor system. In addition, reactivity feedback models appropriate for MSRs were introduced. These developments were verified in the previous studies [18, 19].

While verification demonstrates that the governing equations are correctly implemented and numerically solved, validation against experimental data is essential to assess the predictive capability of a code for real reactor behavior. In this study, the extended MARS code is assessed against three transient experiments - the pump startup test, pump coastdown test, and natural circulation test - of the Molten Salt Reactor Experiment (MSRE) [20–22], which operated at Oak Ridge National Laboratory (ORNL) in the 1960s and provided a unique and invaluable experimental database.

The objective of the present study is to benchmark the extended MARS code to assess its capability to predict key MSR-specific phenomena and limitations. Particular emphasis is placed on evaluating the code's ability to model the drift of DNPs and the resulting impact on reactor kinetics, as well as the calculation of reactivity feedback under natural circulation conditions. By comparing the MARS simulation results with experimental observations, this study aims to provide quantitative evidence of its applicability to MSR transient analysis.

This paper is organized as follows. Section 2 briefly presents the modified PKM of MARS for MSR applications. Section 3 describes the MSRE facility and the MARS modeling for the selected tests. Section 4 discusses the results of the benchmark calculations and compares them with experimental data. Conclusions are drawn in the last section.

2. Modified point kinetics model of the MARS code for MSR applications

The MARS code is a thermal-hydraulic system code that incorporates a built-in PKM to simulate the dynamic behavior of nuclear reactors [13]. It is also coupled with a three-dimensional core neutronics code through a dynamic link library interface [14,15]. Recently, the existing PKM has been modified for applications to liquid-fueled MSRs [19]. In this section, the modified PKM is briefly introduced.

2.1. Modified point kinetics model

The MARS code employs a PKM originally developed for solid-fueled reactors. However, for liquid-fueled MSRs, the PKM must account for the flow effects of DNPs. In earlier PKM formulations for MSRs [23,24], the neutron and DNP population balance equations were expressed as follows:

$$\frac{dN(t)}{dt} = \frac{\rho(t) - \beta_{eff}}{\Lambda} N(t) + \sum_{i=1}^{NG} \lambda_i C_i(t), \tag{1}$$

$$\frac{dC_i(t)}{dt} = \frac{\beta_i}{\Lambda} N(t) - \lambda_i C_i(t) - \frac{C_i(t)}{\tau_C} + \frac{C_i(t-\tau_L)}{\tau_C} e^{-\lambda_i \tau_L}, \tag{2}$$

where β_{eff} is the effective fraction of delayed neutrons that remains in the reactor core, accounting for the loss of DNPs due to fuel flow fraction. τ_C and τ_L are the fuel transit times in the core and the external loop, respectively. The third and fourth terms on the right-hand side of Eq. (2) were introduced to account for the flow effects. However, this approach is only valid when the reactor system consists solely of the core and a single external loop. In a realistic MSR system, additional bypass flow paths may exist, such as those associated with chemical processing or helium bubbling. In such cases, accurately defining the fuel transit time in the external loop becomes challenging, rendering the direct application of Eq. (2) impractical. Furthermore, Eq. (2) is unable to accurately capture reactivity variations associated with flow transients, such as pump start-up conditions [25].

These limitations can be effectively overcome by solving a DNP transport equation over the entire reactor system, including the core and all directly connected fluid systems, rather than relying on Eq. (2):

$$\frac{\partial c_i(r,t)}{\partial t} + \nabla \cdot \left(c_i\,(r,t) \nu_f(r,t)\right) = f \frac{\beta_i}{\Lambda} n(r,t) - \lambda_i c_i(r,t), \tag{3}$$

where $c_i\,(r,t)$ and $n\,(r,t)$ are the DNP and neutron densities at position $\boldsymbol{r}$ at time t, respectively. $\nu_f(r,t)$ is the fuel salt velocity. The coefficient f is one within the reactor core and becomes zero outside of the reactor core. The first term in the right-hand side of Eq. (3) means the DNP generation by nuclear fissions. The detailed representation of this term differs slightly from one researcher to another [12,26,27]. The formulation of Eq. (3) is based on the assumptions that the properties of the molten salt are not affected by the presence of DNPs, and that DNPs are transported solely by the molten salt at its velocity. The diffusion term was omitted at this stage because its effect is considered negligible [12,28].

In the MARS code, the governing equations for two-phase flow are solved numerically over the flow system. The equation set is formulated to simulate single-phase liquid, single-phase gas, or two-phase mixtures within each computational cell. It consists of the continuity, momentum, and energy equations for both liquid and vapor phases, along with a mass conservation equation for non-condensable gases, resulting in a total of seven equations for the fluid system. However, in this study, only the single-phase liquid formulation is employed.

After a timestep advancement in the MARS code, the resulting liquid-phase velocity $\nu_f(r,t)$ is transferred to Eq. (3) and, then, it is solved using the semi-implicit, first-order upwind, finite difference scheme [13,19] to obtain the DNP density at each computational cell. Thereafter, the effective number of *i*-th group DNPs within the reactor core, $C_i(t)$, is determined by:

$$C_i(t) = \left(\sum_{k=1}^{NC} \phi_k^* c_{i,k}(t) V_k \Bigg/ \sum_{k=1}^{NC} \phi_k^* \phi_k V_k \right) \cdot \sum_{k=1}^{NC} V_k, \tag{4}$$

where ϕ_k^* is the importance [29,30] and ϕ_k is the normalized flux. Both ϕ_k^* and ϕ_k are user-specified inputs in this model. *NC* denotes the number of thermal-hydraulic computational cells for the reactor core. $C_i(t)$ is substituted to Eq. (1) and, then, it is solved by the 4th-order Runge-Kutta method to obtain the transient neutron population $N(t)$.

Consequently, Eqs. (1), (3) and (4) replace the earlier PKM formulation based on Eqs. (1) and (2). The advantages of this modified PKM have been demonstrated in previous studies [12,19]. By using Eq. (3), it becomes possible to analyze reactors with complex system configurations, unlike the earlier point kinetics approach and to accurately couple the model with one- or three-dimensional reactor kinetics models.

The transient reactivity $\rho(t)$ in Eq. (1) incorporates the effects of control rods, poisons, and the feedback from fuel temperature, moderator temperature, reflector temperature, etc.:

$$\rho(t) = \rho_{rod} + \rho_{poison} + \alpha_{FT}(T_F - T_{Fo}) + \alpha_{MT}(T_M - T_{Mo}) + \alpha_{RT}(T_R - T_{Ro}), \tag{5}$$

where α is a reactivity coefficient. All temperatures in Eq. (5) are defined as flux-weighted averages to account for neutron importance. However, this is not a unique choice; alternative approaches, such as weighting by the square of the flux, may also be introduced [30].

2.2. Initial steady-state condition and effective delayed neutron fraction

To perform a transient analysis of a thermal-hydraulic system, appropriate initial conditions are required. In practice, these initial conditions correspond to a steady-state condition.

To obtain an initial steady state for an MSR system, a constant core power with a user-specified profile is assumed and null transient advancements for the fluid flow and DNP transport are carried out. This means that Eq. (1) is not solved for a steady-state calculation. When the flows in the entire thermal-hydraulic system reach a steady state, the distributions of DNPs in the reactor system are also determined by Eq. (3). Then, the steady-state DNPs in the reactor core, C_{io}, can be obtained using Eq. (4). Since dN/dt and $\rho(t)$ in Eq. (1) become zero in a steady state, the effective delayed neutron fraction β_{eff} can be obtained as:

$$\beta_{eff} = \frac{\Lambda}{N_o}\sum_{i=1}^{6}\lambda_i C_{io}. \tag{6}$$

where N_o is the steady-state neutron population in the reactor core. The resulting β_{eff} is then applied in subsequent transient calculations.

3. The MARS input model for the MSRE simulation

This section presents the MSRE facility, the selected experiments, and the MARS modeling that includes the thermal-hydraulic input model and point kinetics parameters for the simulations of the selected MSRE tests.

3.1. The MSRE facility and the selected tests

The MSRE was a graphite-moderated, liquid-fueled experimental reactor operated at ORNL from 1965 to 1969 [20,21]. The primary objective of the MSRE program was to demonstrate the technical feasibility and operational characteristics of MSRs, including fuel salt circulation, heat removal, and reactor kinetics behavior. During its operation, the MSRE generated the only integral experimental data set obtained from a reactor-scale liquid-fueled MSR. This unique database remains highly valuable for the validation and benchmarking of system analysis codes for MSR applications.

The MSRE mainly consists of a primary reactor system and a secondary cooling system. As shown in Fig. 1, the primary system consists of the reactor vessel, a centrifugal pump, the shell side of a tube-bundle heat exchanger, and other pipes. The reactor core was housed within a vessel containing a graphite block moderator penetrated by approximately 1140 vertical flow channels, through which the molten salt fuel circulated.

During normal operation, the centrifugal fuel pump provided forced circulation of the fuel salt through the tube-bundle heat exchanger, reactor inlet pipe, flow distributor, downcomer, lower plenum, reactor core, upper plenum, and reactor outlet pipe. Heat generated by nuclear fission in the reactor core was transferred to the secondary system through the shell side of the tube-bundle heat exchanger.

The secondary cooling system removed heat from the primary system via the tube side of the heat exchanger, and the heat was ultimately rejected to the atmosphere through an air-cooled radiator. It is known that although the reactor was originally designed to operate at a thermal power of 10 MW, the maximum operating power was limited to 8 MW by the capacity of the air-cooled radiator system [31,32].

In addition, fuel drain tanks were installed below the reactor vessel as a passive safety feature, allowing the molten salt fuel to be drained from the core under shutdown or abnormal conditions. However, the tanks were not involved in the transient tests analyzed in the present study and are therefore not modeled.

Throughout the MSRE experimental program, a wide range of steady-state and transient tests were conducted, providing comprehensive information on the coupled neutronic and thermal-hydraulic behavior of a liquid-fueled reactor. Among the MSRE experiments, three tests were selected in this study; pump startup and coastdown tests [22] and a natural circulation test [33].

The pump startup and coastdown tests are particularly well suited for benchmarking reactor kinetics models that account for fuel flow effects. These tests were conducted under nearly zero-power conditions and

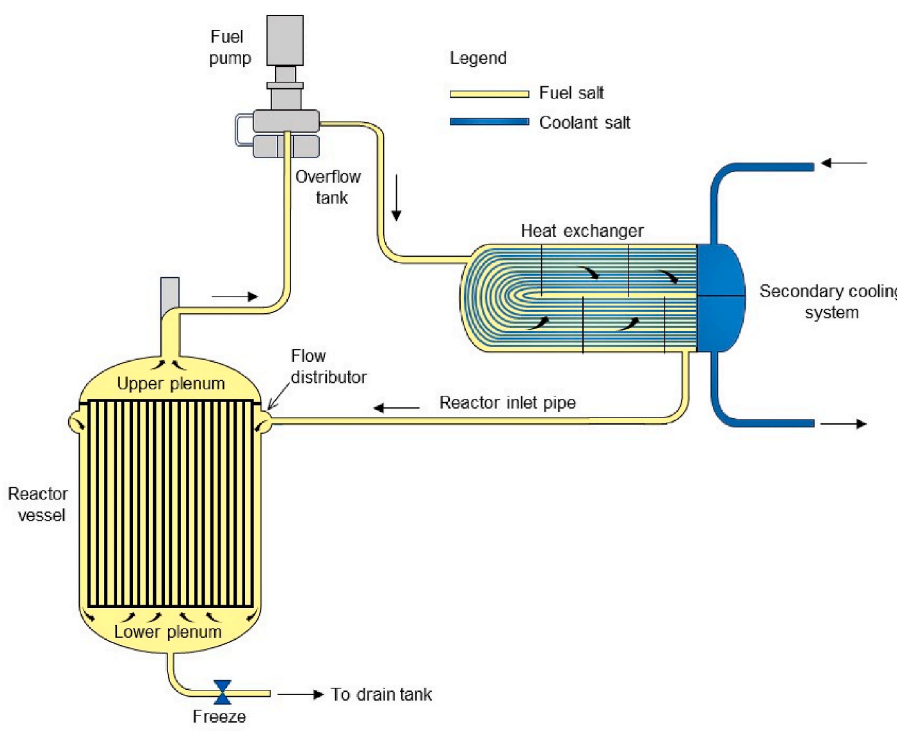

Fig. 1. The schematic of the MSRE to be simulated by MARS.

were designed to separately examine the effect of fuel flow on reactor kinetics. The natural circulation test provides valuable information on the combined effects of DNP drift and reactivity feedback mechanisms. In this test, the reactor power is initially maintained near zero and the fuel salt circulation pump is kept stopped. Then the heat removal by the secondary cooling system is increased stepwise. This induces temperature change of the fuel salt, inducing natural circulation in the primary system, reactivity feedback, and power change. The test procedures are described in more detail in Section 4.

3.2. The MSRE system modeling

Based on the characteristics of the MSRE facility [20] and the selected tests, a one-dimensional, thermal-hydraulic input model for MARS was developed. The MARS nodalization is shown in Fig. 2, while the initial and boundary conditions for each test are described in Section 4.

As shown in Fig. 2, the MSRE primary and secondary systems were modeled using a total of 387 control volumes and 400 junctions. To focus on the behavior of interest, the simulation domain was limited by imposing boundary conditions at the inlet and outlet of the secondary side of tube-bundle heat exchanger.

To model the flow channels in the reactor core, the 1140 vertical channels were grouped into 15 concentric radial rings, and each ring was subdivided into 20 axial nodes. As a result, the reactor core region was represented by a total of 300 thermal-hydraulic computational cells. The lower and upper plena of the reactor vessel were each modeled with three axial nodes.

Unlike solid-fueled reactors, it is not straightforward to precisely define the reactor core region in the MSRE. Thermal neutrons leaking from the graphite moderator region can induce fission reactions not only within the nominal core region but also in the lower and upper plena of the reactor vessel, and even partially in the downcomer [9]. Although a neutron importance cut-off enhances physical consistency in defining the core, the lack of a standardized method for selecting the threshold value introduces a degree of ambiguity into the model. In this study, portions of the lower and upper plena (Volumes 120-03 and 190-01 in Fig. 2) were included in the reactor core and defined as the lower and upper axial boundaries, respectively. The axial power profile in the reactor core was assumed to be a cosine shape, as shown in Fig. 3, and the radial power profile was taken from the Serpent calculation results [9]. This power shape is assumed to remain unchanged during the transients. The blue dotted line in Fig. 3 represents the axial power profile used for a sensitivity calculation (see Section 4.1). The importance ϕ_k^* in Eq. (4) is assumed to be unity for all the calculations in Section 4.

A flow distributor is installed at the upper part of the reactor vessel and is designed to distribute the flow uniformly into the downcomer. However, multi-dimensional flow is likely to develop within the downcomer region. Nevertheless, in the MARS input model, this region

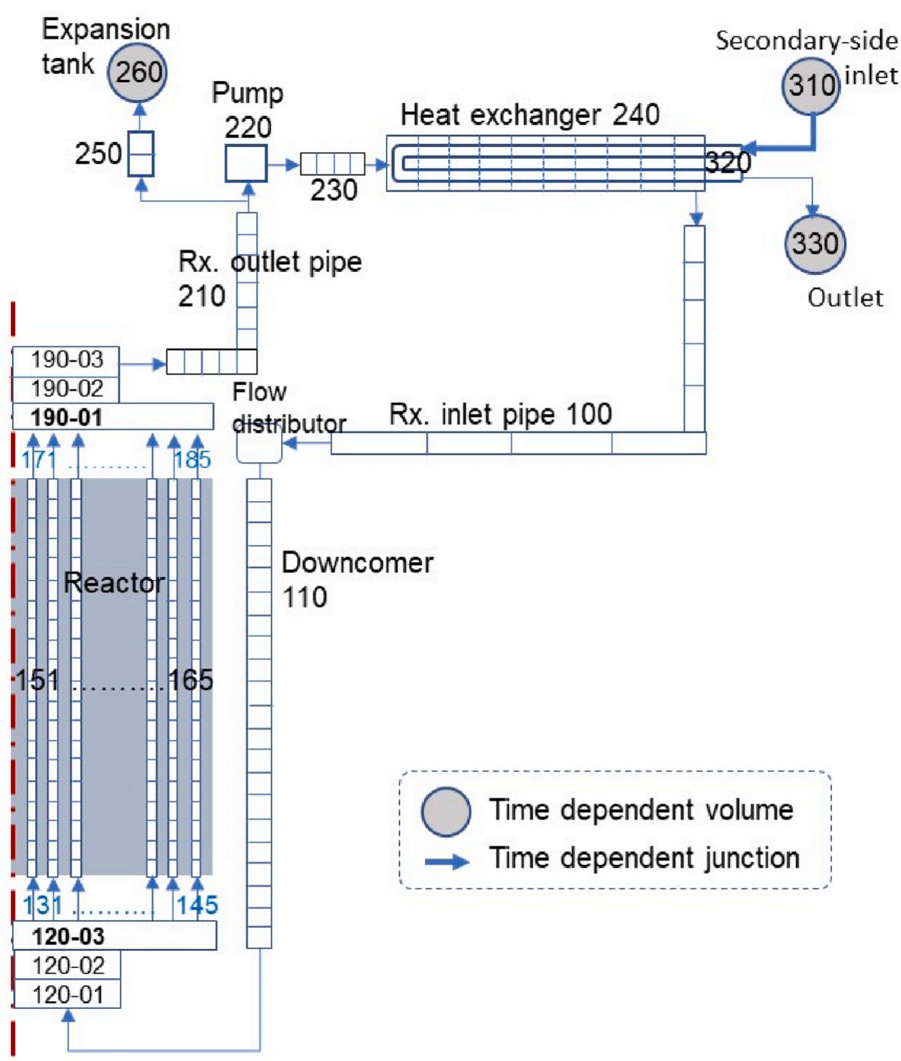

Fig. 2. The MSRE nodalization for the MARS code.

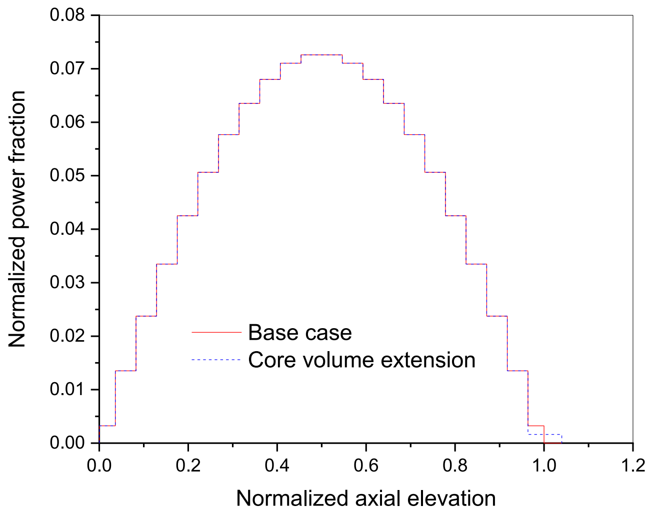

Fig. 3. Comparison of axial power profiles (user-specified).

was modeled using vertical one-dimensional meshes.

The fuel pump was simulated using the built-in centrifugal pump model in the MARS code. Since detailed information on the pump head and torque characteristics and the moment of inertia was not available, typical generic pump parameters were adopted.

The MSRE heat exchanger is a tube-bundle type with baffles, in which the fuel salt flows on the shell side and the coolant salt flows on the tube side. The shell side was divided into 10 vol, and the tube side was divided into 20 volumes. The tube walls were modeled as heat structures to simulate heat transfer between the primary and secondary systems. For the shell side, the cross-flow heat transfer option in the MARS code was employed to appropriately represent the heat transfer characteristics of the baffled tube-bundle configuration.

The expansion tank, Component 260 in Fig. 2, was included to accommodate thermal expansion of the primary fuel salt and to define the system pressure. It was modeled as a time-dependent volume.

The measured coolant salt flow rate and temperature at the heat exchanger inlet in the secondary system were imposed as boundary conditions, while the pressure was specified at the outlet of the heat exchanger (Component 330).

The graphite moderator blocks in the reactor vessel were modeled as heat structures. Although a non-negligible fraction of the fission energy is directly deposited in the graphite, all fission power was assumed to be deposited in the fuel salt as a volumetric heat source due to current code limitations. In reality, heat generated within the graphite is transferred to the fuel salt under steady-state conditions. By contrast, assuming that all heat is deposited in the fuel salt leads to deviations from the actual physical behavior. Under this assumption, the fuel temperature responds more rapidly, while the graphite acts as a passive heat reservoir with a slower thermal response, resulting in an underestimation of the average graphite temperature. This modeling assumption should be improved in future work, particularly for simulations of high-power conditions. However, its impact on the three tests analyzed in this study is expected to be not significant, as the pump tests are conducted under nearly zero power conditions and the natural circulation test under low power conditions. A detailed discussion is provided in Section 4.3.

The walls of the reactor vessel, pump and other pipes are assumed to be insulated. Some uncertain input parameters, including the form loss factors in the primary system and the heat transfer area of the heat exchanger, were calibrated to reproduce a steady-state full-power condition [32]. The calculated fuel salt transit time of the reactor system under full-power condition was 25.63 s, whereas the experimentally reported value [27,34] is 25.2 s.

3.3. Reactor kinetics parameters

All the reactor kinetics parameters of the MSRE were obtained from Ref. [9] to maintain consistency. The 6-group delayed neutron parameters are listed in Tables 1 and 2. It is noted that U-235 was used for the pump startup and coastdown tests, whereas U-233 was used for the natural circulation test. The corresponding prompt neutron generation time and reactivity coefficients are listed in Table 3.

4. The results of the transient simulations

This section briefly introduces the procedures of the pump startup, coastdown, and natural circulation tests and presents the results obtained from the MARS simulations. The discussion focuses on the key phenomena observed in each test.

4.1. The pump startup test

The pump startup test [22] was performed under near zero-power conditions, with the reactor maintained at a very low power level of approximately 100 W to suppress thermal feedback and a temperature of 908 K. The primary fuel pump and secondary coolant pumps were initially at rest, resulting in stagnant molten salt in both the primary and secondary systems. The transient test begins with the startup of the fuel pump, followed by the startup of the coolant pump 1 s later. When the fuel pump was started, the flow rate was increased to its nominal value in about 10 s.

Under the condition of nearly zero power, thermal feedback effects are negligible, and changes in reactivity are primarily governed by the DNP drift effect. As the fuel circulation begins, the DNPs generated in the reactor core are transported out of the core region and decay in the external loop, leading to a reduction in the effective delayed neutron fraction. This introduces a negative reactivity and causes a decrease in reactor power. In the test, the reactivity effects of fuel flow-rate transient were measured by letting the flux servo controller hold the reactor critical during the transient; that is, the reactivity added by the control rod is then equal and opposite to the reactivity change due to the flow transient. The servo controller movement was controlled by three signals: the measured flux, the flux demand (a constant flux in this test), and the rod speed signal.

In the MARS simulation, the control rod operation was not modeled directly, as the control logic is not known in detail. Instead, the ideal transient reactivity due to control rod motion was derived from Eq. (1), assuming perfect control without temperature feedback; that is, $dN(t)/dt$ in Eq. (1) is zero and, thus, the reactivity by control rod motion can be obtained as follows:

$$\rho(t) = \beta_{eff} - \frac{\Lambda}{N(t)} \sum_{i=1}^{NG} \lambda_i C_i(t). \tag{7}$$

Eq. (7) is inserted to Eq. (1) to compensate the reactivity change induced by the fuel salt flow. Fei et al. [35] also adopted the same approach.

4.1.1. Transient behaviors

The fuel salt flow rate during pump start-up rises to the rated value in

Table 1
The delayed neutron parameters of U-235 [9].

Group	Fraction	Decay constant (1/s)	Half-life (s)
1	0.000208	0.0125	55.452
2	0.001190	0.0318	21.797
3	0.001070	0.1090	6.3591
4	0.003020	0.3170	2.1866
5	0.000948	1.3500	0.5134
6	0.000345	8.6400	0.0802

Table 2
The delayed neutron parameters of U-233 [9].

Group	Fraction	Decay constant (1/s)	Half-life (s)
1	0.000208	0.0125	55.452
2	0.000830	0.0323	21.460
3	0.000595	0.1050	6.6014
4	0.001200	0.2940	2.3576
5	0.000226	1.2400	0.5590
6	0.000192	10.020	0.0680

Table 3
The prompt neutron generation time and reactivity coefficients [9].

Parameter	U-235	U-233
Prompt neutron generation time (s)	2.4×10^{-4}	4.0×10^{-4}
Fuel temp. coefficient (pcm/K)	−8.71	−11.3
Graphite temp. coefficient (pcm/K)	−6.66	−5.81

nearly 10 s as can be seen in Fig. 4. It is noted the fuel salt flow rate was not directly measured but estimated from the pump speed [34]. The maximum standard deviations of the normalized flow rate were reported to be 0.031 and 0.022 for the pump startup and coastdown tests, respectively. The rate at which the pump flow increased was somewhat slow in the base case calculation. In the MARS code, the pump speed is determined by solving an angular momentum equation for a centrifugal pump, in which key parameters include motor torque, pump hydraulic and frictional torque, and moment of inertia. A sensitivity calculation was performed with the pump moment of inertia reduced by 50%. In this case, the calculated flow rate agrees very well with the estimated value, as illustrated in Fig. 4.

Fig. 5 shows the transient control rod reactivity. As the stagnant fuel salt begins to move, the loss of delayed neutrons starts, which leads to the withdrawal of the control rods to maintain criticality. As can be seen in Fig. 5, the control rod reactivity becomes relatively stable between 25 s and 45 s. During this period, the average values from the experiment and the MARS calculation are 227.3 pcm and 222.4 pcm, respectively. The results are in reasonably good agreement.

When the DNPs that have flowed from the reactor core into the loop re-enter the core, a positive reactivity insertion occurs. Consequently, the control rod withdrawal must be stopped and reversed to insertion. The repetition of this phenomenon leads to oscillations in reactivity. These oscillations can be clearly explained by the initial DNP distribution and the fuel salt transit time through the reactor system:

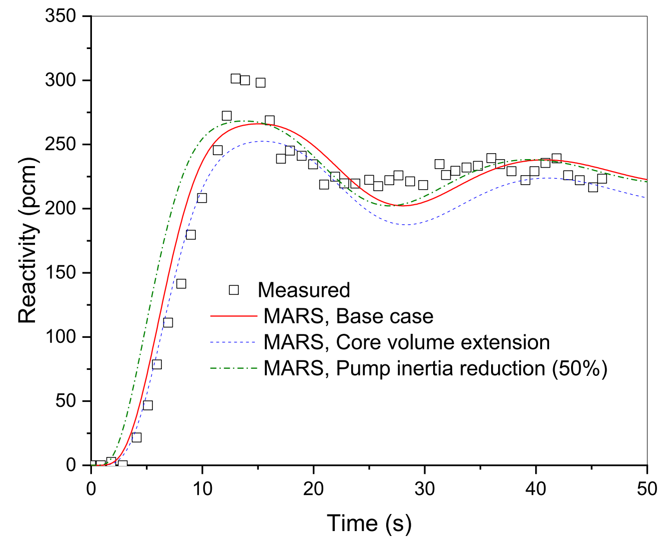

Fig. 5. Transient control rod reactivity of the pump startup test.

- At the beginning of the test, DNPs exist only in the reactor core, while those in the external loop have completely decayed out.
- Once the fuel pump begins to operate, the long-lived DNPs initially generated in the core periodically re-enter the core after one fuel salt transit time, inducing oscillations that gradually decay over time.

Fig. 6 shows the reactivity variation of the base case over 150 s. The first three peaks of the reactivity occur at 15.3 s, 40.7 s, and 66.2 s, yielding a period of about 25.5 s. This period agrees well with the reactor system transit time. MARS can realistically reproduce the transient characteristics of the reactivity. However, since control rod actions are not modeled in the MARS calculation, the reactivity overshoot observed in the experiment around 15 s does not appear.

When the pump moment of inertia was reduced, the transient reactivity response proceeds faster than in the base case, as expected. Nevertheless, the overall magnitude of the reactivity change at around 50 s remains almost identical, as shown in Fig. 5, since no other reactivity change mechanisms are involved.

4.1.2. Asymptotic behavior of the reactivity loss induced by fuel flow

In the pump startup test, the ultimate reactivity effect is determined

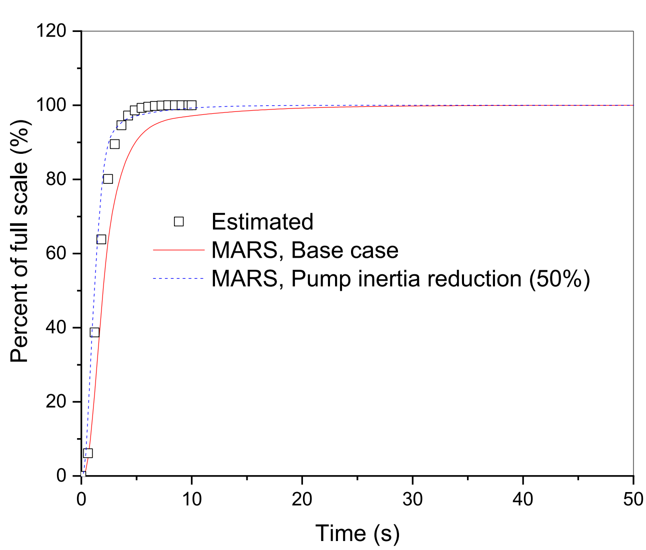

Fig. 4. Transient fuel salt flow rate of the pump startup test.

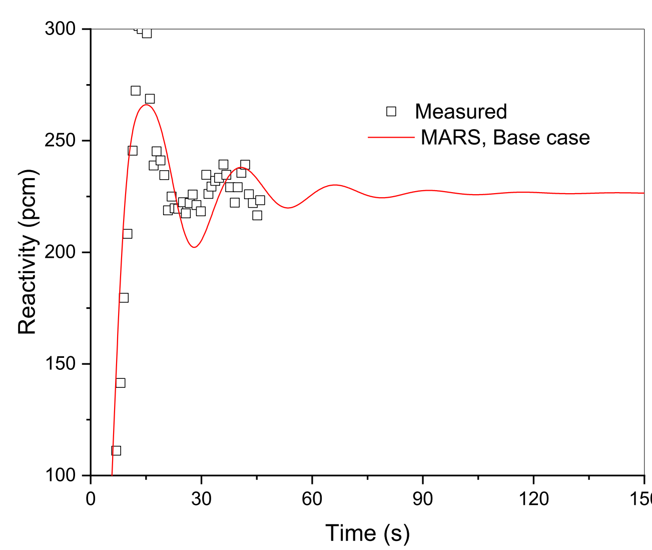

Fig. 6. Transient control rod reactivity of the pump startup test during 150 s.

solely from the loss of DNPs associated with fuel salt flow. The primary system of MSRE can be simply divided into the reactor core and the external loop. In this configuration, the steady-state reactivity loss due to flow can be evaluated analytically with a uniform importance [36, 37]. For example, assuming a cosine-shaped axial power profile in the reactor core, Eq. (3) under steady-state conditions is integrated by first assuming the DNP density at the reactor inlet. It is integrated from the core inlet to the core outlet and then from the loop inlet to the loop outlet (i.e., back to the reactor inlet). The DNP density obtained at the loop outlet is equal to the initially assumed value at the reactor inlet. Using this relationship, the DNP density distribution can be determined and, in turn, the reactivity loss due to DNP drift can then be derived as follows:

$$\Delta\rho_{Drift} = \sum_{i=1}^{NG}\left[\frac{\beta_i}{2}\frac{(1+e^{-\lambda_i\tau_C})(1-e^{-\lambda_i\tau_L})}{(1-e^{-\lambda_i(\tau_C+\tau_L)})\left(1+(\lambda_i\tau_C/\pi)^2\right)}\right], \tag{8}$$

where τ_C and τ_L are the fuel salt transit times through the reactor core and the external loop, respectively. Eq. (8) indicates that the reactivity loss depends exclusively on τ_C and τ_L under the assumed power profile and the conditions of uniform importance and no flow mixing. When a non-uniform importance distribution is considered, the reactivity loss may change accordingly.

In the MARS results at 150 s, τ_C and τ_L were 9.56 s and 16.14 s, respectively. The asymptotic reactivity loss shown in Fig. 6 is 226.5 pcm, whereas the value calculated using Eq. (8) is 228.4 pcm. The results show very good agreement. This result was obtained using a uniform importance; however, the good agreement does not imply that this assumption is appropriate. Rather, the result appears to be influenced by an error compensation effect.

It is clear that, if τ_C is increased a little while τ_L is reduced by the same amount, the delayed neutron loss decreases. To examine this effect quantitatively, the axial length of Component 190-01 in Fig. 2 was arbitrarily extended from 0.0635 m to 0.1435 m, while the same length was reduced from Components 190-02 and 190-03; that is, the core volume was extended. As a result, the core transit time increased by 1.11 s, while the loop transit time decreased by the same amount. Fig. 5 shows that, in the case of core volume extension, the reactivity loss is reduced by 14.2 pcm at 50 s. This again confirms the effects of τ_C and τ_L on the DNP drift effect.

4.2. The pump coastdown test

The pump coastdown test [22] can be regarded as the inverse of the pump startup test. Similar to the startup test, it was conducted at a very low power level of approximately 100 W and a temperature of 908 K. The primary fuel circulation pump was initially operating at a rated flow rate of 168 kg/s and was then tripped, allowing the flow rate to decrease gradually due to pump inertia. As the fuel salt flow decreases, the DNPs that were previously transported out of the reactor core remain longer within the core. This leads to an increase in the effective delayed neutron fraction and introduces a positive reactivity insertion. Ultimately, the net loss of delayed neutrons approaches zero. As in the pump startup test, the control rod was adjusted to maintain reactor criticality and was eventually fixed at its equilibrium position.

In this test, the transient progression is relatively simple, because the DNP loss effect gradually diminishes following the pump trip. As in Section 4.1, the control rod motion was not modeled explicitly in the MARS simulation; instead, the reactivity associated with the control rod operation was calculated using Eq. (7).

Successful simulation of this test requires accurate representation of both the transient flow behavior and its coupling to reactor kinetics. The two sensitivity cases performed for the pump startup test were also examined here. The MARS results are shown in Figs. 7 and 8.

Fig. 7 compares the transient fuel salt flow rates with the estimated experimental data [34], of which standard deviation is 1.4% at 20 s. The flow rate approaches zero as time progresses, meanwhile the estimated data are available only up to 20 s. When the pump moment of inertia was reduced by half, the flow rate decreased more rapidly, as expected (see Fig. 7). Although the control rod reactivity appears slightly earlier (see Fig. 8), the overall trend becomes increasingly similar over time. In the pump startup test, reducing the moment of inertia led to better agreement with the experimental data; however, in this case, it resulted in larger discrepancies. This indicates that modifying the moment of inertia alone is insufficient to obtain improved results, and that the hydraulic and frictional torques of the pump should also be adjusted simultaneously. Since detailed information on these parameters is not available, no further attempts were made to improve the results.

It is noted that the long-lived DNPs require more time to reach equilibrium; however, as shown in Fig. 8, the measurement duration is too short to observe the equilibrium reactivity. The reactivity variation up to 70 s is in good agreement.

When the reactor volume is extended, the DNP residence time in the core also increases, thereby reducing the DNP loss under the initial steady-state condition. Consequently, the final reactivity increase following the pump trip is correspondingly reduced. Fig. 8 illustrates

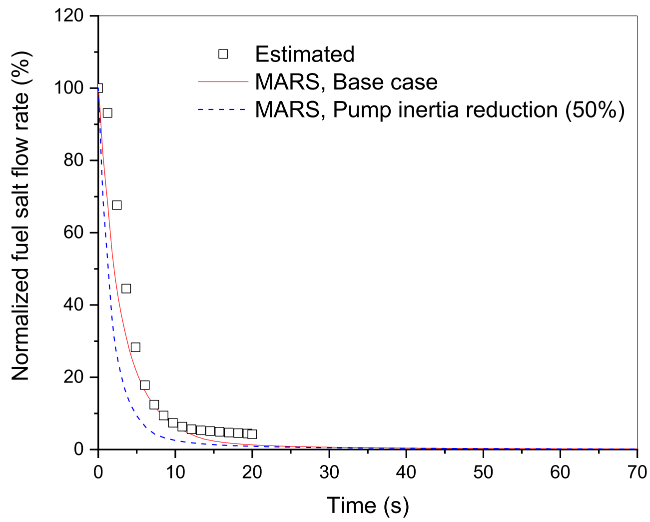

Fig. 7. Transient fuel salt flow rate of the pump coastdown test.

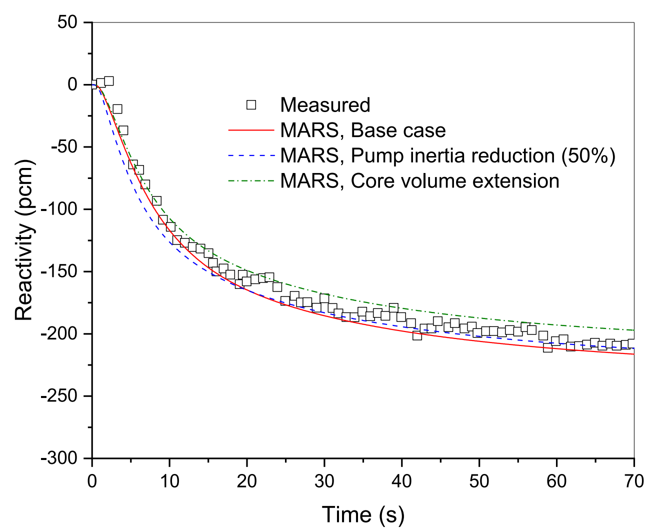

Fig. 8. Transient control rod reactivity of the pump coastdown test.

this behavior.

4.3. The natural circulation test

The natural circulation test [33] represents a more complex transient scenario, involving strongly coupled neutronic and thermal-hydraulic effects. The key test conditions can be summarized as follows:

- The reactor was operated with U-233 fuel salt with the primary fuel pump turned off. The initial fuel salt flow rate was not known in the test; however, the MARS analysis predicts a value of approximately 1.46 kg/s, which corresponds to a very low velocity of about 1–2 mm/s within the reactor core flow channels.
- The initial reactor power was set to be 8.0 kW [10], assuming the reactor was at steady-state equilibrium condition. In reality, the reactor does not appear to have reached thermal-hydraulic equilibrium during the early stage of the test; this issue is discussed later.
- The control rods remained in its position during the transient.
- The secondary coolant flow was maintained constant at the rated flow rate during the transient; however, heat removal through the radiator increased in a stepwise manner at 0, 1,200, 4,500, 8,400, and 11,700 s.

In the MARS calculation, the transient coolant salt temperature at the secondary-side inlet of the tube bundle heat exchanger, retrieved from Ref. [33], was provided as a boundary condition. The enhanced heat removal reduced the fuel salt temperature, thereby inducing natural circulation of the fuel salt. Under these conditions, both the fuel salt flow rate and the temperature distribution in the primary loop vary, leading to simultaneous effects from DNP drift and reactivity feedback associated with the changes of fuel salt and graphite temperature. Consequently, this experiment provides a stringent benchmark for assessing the coupled thermal-hydraulic and neutronic modeling capability applied to MSRs.

The lines labeled “MARS BC” in Figs. 9–11 represent the calculation results of the base case (BC). The fuel salt temperature behavior is shown in Fig. 9. It is noted that initial conditions in the test and calculations are significantly different each other. It seems that the initial condition of the test does not reach a thermal-hydraulic equilibrium state [35, 38–40]. In the MARS calculation, the primary-side flow and temperature at the steady state were obtained by null-transient advancement of 5000 s with the fixed secondary-side boundary conditions, resulting a natural circulation with a flow rate of 1.46 kg/s. The initial fuel salt temperatures of MARS are in good agreement with the energy balance of the

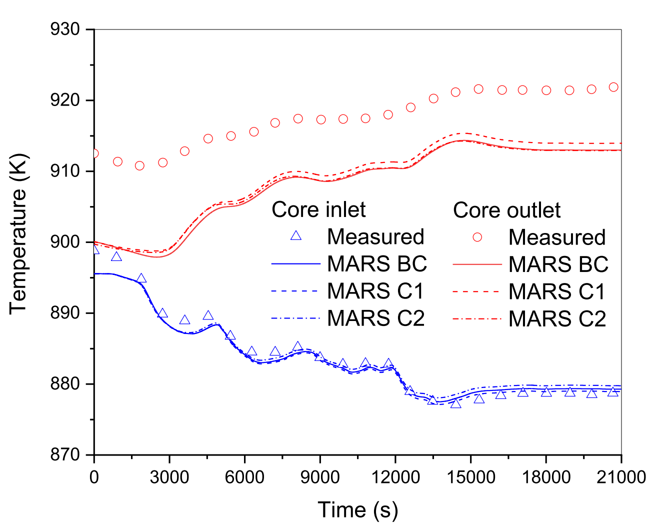

Fig. 9. Fuel salt temperature behaviors of the natural circulation test.

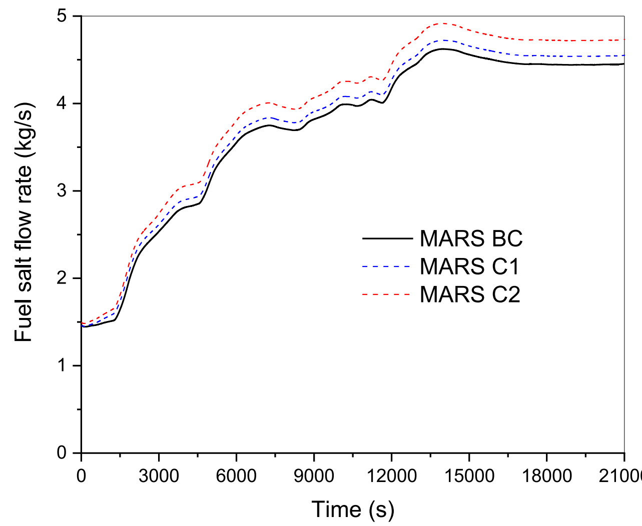

Fig. 10. Fuel salt flow rate behaviors of the natural circulation test.

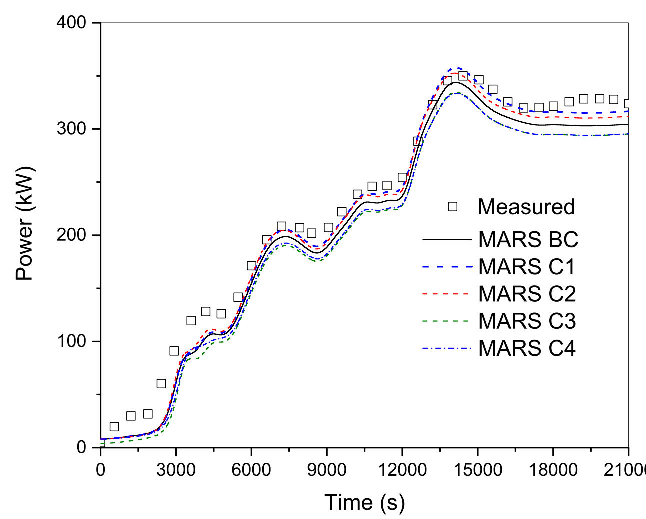

Fig. 11. Transient core power during the natural circulation test.

reactor power. However, it is difficult to verify whether the overall energy balance is satisfied for the measured temperatures, since the initial fuel salt flow rate was not reported in the experiment. Similar differences in the initial conditions were also observed in calculations performed with the SAM and SPECTRA codes [38].

It should be noted that, during the transient, the reactivity change induced by temperature variations is determined by the temperature deviation from the initial condition, as expressed in Eq. (5). Therefore, the absolute value of the initial temperature itself does not significantly affect the calculated reactivity response. As shown in Fig. 9, although there is a considerable difference between the experimental and calculated fuel salt temperatures at the reactor outlet, the overall trends of the transient behavior in Figs. 9 and 11 are quite similar.

Since the power variation is mainly driven by changes in fuel salt temperature and flow rate, their relative effects are first briefly evaluated. Fig. 10 shows that the flow rates in the base case calculation are 1.46 kg/s at 0 s and 4.45 kg/s at 21,000 s. Based on these values, the core and loop transit times can be approximately estimated and substituted into Eq. (8), yielding steady-state reactivity losses of 0.9 pcm at the beginning and 6.7 pcm at the end. Thus, an additional reactivity loss of about 5.8 pcm can occur due to the flow effect during the transient.

Meanwhile, the average fuel salt temperature decrease in the primary system reaches approximately 4.0 K. Considering the reactivity coefficients listed in Table 3, the corresponding positive reactivity amounts to around 60 pcm. Therefore, the power change observed in the test can be regarded as being predominantly driven by the fuel salt temperature variation [33]. In this regard, a total of four sensitivity cases were evaluated:

C1: 10% heat exchanger area increase,
C2: Form loss reduction at the fuel pump inlet and exit (from 1.75 to 0.5),
C3: Initial core power of 4 kW (8 kW at the base case),
C4: Graphite moderator temperature coefficient, 0.0 pcm/K.

In the base case in Fig. 11, the reactor power shows a significant discrepancy from the experimental data in the early stage, which appears to be due to differences in the initial conditions. The subsequent power behavior is qualitatively well predicted; however, the power is consistently underestimated. At 21,000 s, a value of 304.5 kW is predicted, which is lower than the experimental value of about 354 kW.

By increasing the heat transfer area of the heat exchanger (Case C1), the core inlet fuel salt temperature decreases slightly due to enhanced heat removal, leading to a positive reactivity insertion. Compared with the base case, the final reactor power increases by 12.2 kW. With the increase in power, the core outlet temperature rises by a smaller amount than the decrease in the core inlet temperature. As a result, the overall power increase is limited.

It is noted that the MSRE heat exchanger is a baffled tube-bundle heat exchanger, with the fuel salt flowing on the shell side. The baffles installed on the shell side induce significant cross-flow and thereby enhance heat transfer. Typically, the shell-side heat transfer coefficient increases by a factor of approximately 2-5 compared to longitudinal (no-baffle) flow, and the overall heat transfer rate may increase by roughly 30-100% when segmental baffles are used. As a result, the uncertainty of the heat transfer model can be significant. Moreover, under natural circulation conditions, where the flow rate is much lower than the nominal value (approximately 170 kg/s), the uncertainty in the shell-side heat transfer coefficient becomes even larger. Therefore, the heat exchanger model can be regarded as a determining factor in the present calculation results.

When the form loss factors at the fuel pump are reduced, the flow rate increases significantly, as shown in Fig. 10, which in turn enhances the heat transfer to the secondary side. Accordingly, the final reactor power becomes the second highest in Fig. 11. Cases 1 and 2 both directly affect heat removal to the secondary side. A difference is observed in the temperature difference between the core inlet and outlet: it increases in Case C1, but decreases in Case C2.

When the initial power is reduced from 8 kW to 4 kW (Case 3), the early transient behavior shows small differences and, as time progresses, the difference in power increases slightly.

Case 4 was designed to indirectly assess the impact of the graphite block modeling issue discussed in Section 3.2 by removing the graphite moderator temperature feedback. The results of Case 4 are compared in Fig. 11, and the overall trend is similar to that of the base case. This indicates the transient is mainly driven by fuel temperature feedback. However, the difference is not negligible, and the core power at 21,000 s is predicted to be the lowest among the cases. This means that accurate modeling of heat generation in the graphite block is more important than initially anticipated and will become even more significant under higher power conditions.

Overall, the results indicate that MARS reasonably captures the qualitative trend of the power variation over time. Heat removal by the secondary side is identified as the dominant factor in this test, and more accurate modeling of this effect leads to improved predictions. The heat exchanger parameters in the base case were adjusted to reproduce full-power operating conditions well; however, the current MARS model under low-flow natural circulation appears to underestimate heat transfer performance. In addition, accurate modeling of the heat source in the graphite block was found to be important and should be treated as a top priority in future code development.

5. Conclusions

In this study, the MARS thermal-hydraulic system code was assessed for applications to liquid-fueled MSRs. To enable MSR analysis, key modifications were incorporated in the MARS code, including the implementations of molten salt thermophysical property models and a DNP transport equation to replace the conventional precursor balance equation used in solid-fueled point kinetics models. In addition, a transport-based decay heat model was incorporated. These features, previously verified, extend the applicability of MARS to circulating-fuel systems while preserving its robustness.

The extended MARS code was validated against three MSRE transients: the pump startup test, pump coastdown test, and natural circulation test. The pump startup and coastdown tests were performed under nearly zero-power conditions, eliminating temperature feedback effects and isolating the influence of DNP drift. The simulations successfully reproduced the essential physics of DNP loss associated with fuel flow. In the startup test, the oscillatory reactivity behavior was explained by the fuel salt transit time and periodic DNP re-entry, and the predicted oscillation period agreed well with the estimated system transit time. The asymptotic reactivity loss was also consistent with analytical predictions. In the coastdown test, the gradual recovery of delayed neutron contribution as flow decreased was captured with satisfactory accuracy, confirming that the modified point kinetics model properly responds to dynamic fuel flow changes.

The natural circulation test involved coupled neutronic and thermal-hydraulic effects. Although discrepancies in absolute fuel salt temperature values were observed, mainly due to uncertain initial conditions, the overall trends of fuel salt temperature and reactor power were reasonably reproduced. Sensitivity analyses indicated that the power variation was dominated by fuel temperature feedback, while DNP drift played a secondary role. They highlighted the importance of secondary-side heat removal and heat exchanger modeling under natural circulation conditions. In addition, accurate modeling of the heat source in the graphite block was found to be very important and should be treated as a top priority in future code development. Without this modeling capability, the applicability of MARS is inevitably limited.

Overall, the benchmark results demonstrate that the extended MARS code can capture the principal MSR-specific phenomena, including DNP transport effect and coupled thermal-hydraulic and neutronic interactions, while maintaining computational efficiency suitable for system-scale analysis. The direction for future code improvements, which involve the refinement of heat exchanger model under natural circulation, heat source model in the graphite block, and neutron importance for the modified point kinetics model, has also become clearer. Moreover, the scope of code validation should be extended to a wider range of conditions.

CRediT authorship contribution statement

Jae Jun Jeong: Writing – original draft, Validation, Investigation, Data curation, Conceptualization. **Yun Je Cho:** Writing – review & editing, Investigation, Funding acquisition. **Hyun Chul Lee:** Writing – review & editing, Methodology, Investigation. **Byongjo Yun:** Writing – review & editing, Investigation, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Nomenclature

c_i	Density of the *i*-th group DNP (#/m^3)
C_i	Effective number of the *i*-th group DNPs in the reactor core (#)
n	Neutron density (#/m^3)
N	Number of neutrons in the reactor core (#)
$\boldsymbol{r}$	position vector (m)
t	Time (s)
T	Temperature (K)
v_f	Liquid-phase velocity (m/s)
V	Volume (m^3)
α	Reactivity coefficient (−)
β_i	Delayed neutron fraction of the *i*-th group DNP (−)
β_{eff}	Effective delayed neutron fraction (−)
λ	Decay constant (1/s)
Λ	Prompt neutron generation time (s)
ρ	Reactivity (−)
τ	Transit time (s)

Subscripts

F	Fuel salt
FT	Fuel salt temperature
M	Moderator
MT	Moderator temperature
i	*i*-th group DNP
k	Volume index
o	Steady state or reference
R	Reflector
RT	Reflector temperature

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